

A LOG-FREE LOWER BOUND FOR IMAGINARY QUADRATIC FIELDS OF 5-RANK AT LEAST TWO

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ABSTRACT. Let $N_{5,2}^-(X)$ denote the number of imaginary quadratic fields K , counted up to $\mathbb{Q}$ -isomorphism, for which $|\text{Disc}(K)| \leq X$ and $\dim_{\mathbb{F}_5} \text{Cl}(K)/5\text{Cl}(K) \geq 2$. Bartz, Levin, and Thamminana proved the lower bound $N_{5,2}^-(X) \gg X^{1/3}/(\log X)^2$. We present a complete human-readable proof candidate for the log-free estimate

$$N_{5,2}^-(X) \gg X^{1/3}.$$

The argument combines their explicit genus-two curve with the strong power-free-value theorem of Stewart–Top, rather than the logarithmically weaker variant used in the Hilbert-irreducibility framework of Kulkarni–Levin. The main points are to preserve one positive congruence cone, extract bounded witnesses for distinct squarefree values, delete a thin set without losing the quadratic main term, and identify the displayed genus-two curve with the model to which the torsion theorem applies. We give the required endpoint-avoidance and square-class injection lemmas, an exact rational isomorphism of the two genus-two models, and a reproducibility statement. The full analytic argument has undergone independent adversarial review, but has not been formalized end to end in a proof assistant; this document should therefore be treated as a pre-submission research draft.

1. INTRODUCTION

For a positive integer m and a finite abelian group A , write

$$\text{rk}_m A = \max\{r : (\mathbb{Z}/m\mathbb{Z})^r \hookrightarrow A\}.$$

When $m = p$ is prime, this is $\dim_{\mathbb{F}_p}(A/pA)$. We study the counting function

$$N_{5,2}^-(X) = \#\left\{ \begin{array}{l} K/\mathbb{Q} \text{ imaginary quadratic, up to } \mathbb{Q}\text{-isomorphism : } \\ |\text{Disc}(K)| \leq X, \quad \text{rk}_5 \text{Cl}(K) \geq 2 \end{array} \right\}. \quad (1)$$

Byeon proved the unconditional estimate $N_{5,2}^-(X) \gg X^{1/4}$ [3]; an earlier result gives a weaker exponent in greater generality [2]. Kulkarni and Levin developed a Hilbert-irreducibility method that converts rational torsion on hyperelliptic Jacobians into quantitative class-group constructions [5]. Bartz, Levin, and Thamminana applied this framework to an explicit genus-two

Date: Research manuscript draft, 14 August 2026.

curve and obtained

$$N_{5,2}^-(X) \gg \frac{X^{1/3}}{(\log X)^2} \quad [1, \text{Theorem 1.3}]. \quad (2)$$

Conditional results of a different shape are known; for example, Chattopadhyay and Saikia obtain, assuming the *abc*-conjecture, an exponent $1/4 - \varepsilon$ when $p = 5$ [4].

The purpose of this paper is to remove the logarithmic loss in (2) without changing the genus-two input.

Theorem 1.1. *There exist constants $c > 0$ and $X_0 > 1$ such that, for every $X \geq X_0$,*

$$N_{5,2}^-(X) \geq cX^{1/3}.$$

The logarithmic factor in (2) comes from the power-free-value estimate used in the general Kulkarni–Levin construction. For the particular binary sextic arising here, the stronger theorem of Stewart and Top [7, Theorem 1] applies. Its proof produces $\gg H^2$ distinct squarefree values from a fixed positive box of side H , whereas the relevant thin set contains only $O(H)$ rational parameters of comparable height. The difference remains quadratic.

There are three bookkeeping issues that must be resolved before this idea becomes a proof. First, a projective change of variable in the Stewart–Top proof may change the positive cone; we avoid it by choosing the Kulkarni–Levin coordinate so that both endpoint coefficients are nonzero. Second, the thin-set estimate counts rational parameters rather than binary form values; even homogeneity gives the required injection. Third, the genus-two model in the torsion construction must be identified with the displayed curve by an exact rational map. These points are treated in Sections 2, 4 and 6.

To the best of our knowledge within a search of OpenAlex, Semantic Scholar, Crossref, arXiv, and the direct citation chain completed on 14 August 2026, no earlier source states the unconditional log-free estimate in Theorem 1.1. The closest unconditional result with the same exponent is (2). This is a scoped literature statement, not a claim that all indexed and unindexed literature has been exhausted.

2. THE FIXED GENUS-TWO CURVE

Bartz, Levin, and Thamminana exhibit the curve

$$C_0 : y^2 = f_0(x) = (5x + 7)(128x^4 + 549x^3 + 1007x^2 + 936x + 368) \quad (3)$$

and obtain the needed rational 5-torsion input from their Theorem 2.1 [1, pp. 4–7]. The following form of that input is all that we use.

Proposition 2.1. *The curve C_0 has genus two, and*

$$\text{rk}_5 \text{Jac}(C_0)(\mathbb{Q})_{\text{tors}} \geq 2.$$

Moreover, after the change of variables $u = 640x$, $v = 640^2y$, it has the monic odd-degree model

$$v^2 = g(u) = (u + 896)(u^4 + 2745u^3 + 3222400u^2 + 1916928000u + 482344960000). \quad (4)$$

The associated projective branch form has irreducible-factor degrees 1, 1, 4 over $\mathbb{Q}$.

Proof. The identity $g(u) = 640^4 f_0(u/640)$ is immediate by expansion. The quartic factor in (4) is irreducible over $\mathbb{Q}$, and g is separable. Together with the branch point at infinity, this gives factor degrees 1, 1, 4.

For the torsion assertion, substitute $(t, u, z) = (2/3, -1/3, 25)$ into the coefficient formulas of [1, Theorem 2.1]. The resulting even model $C_{\text{even}} : y^2 = P(x)$ is mapped to C_0 by

$$x_0 = \frac{2(1 - 2x)}{5x - 1}, \quad y_0 = \frac{59049y}{1024(5x - 1)^3}, \quad (5)$$

with inverse

$$x = \frac{x_0 + 2}{5x_0 + 4}, \quad y = \frac{8192y_0}{2187(5x_0 + 4)^3}. \quad (6)$$

The exact polynomial identity and the two coordinate compositions are recorded in Section A. Thus the theorem-level torsion conclusion for the even model applies to C_0 . The stronger reported computation of the entire torsion subgroup is unnecessary here. $\square$

3. TWO ELEMENTARY BRIDGE LEMMAS

The first lemma lets us preserve the positive cone in the later Stewart–Top count.

Lemma 3.1 (Finite endpoint avoidance). *Fix the bad-prime data M_0 , a shift N_0/M_0 , and the polynomial g in (4). In the proof of [5, Lemma 3.1], the integer N_1 may be chosen sufficiently large and so that*

$$g\left(-\frac{N_0}{M_0} + \frac{1}{N_1}\right) g\left(-\frac{N_0}{M_0} - \frac{1}{N_1}\right) \neq 0. \quad (7)$$

For this choice, the transformed projective branch form has nonzero leading and trailing coefficients.

Proof. The construction of Kulkarni–Levin permits any sufficiently large positive integer N_1 . Each equality obtained by setting one factor in (7) equal to zero excludes only finitely many such integers, because g has finitely many roots. The two values in (7) are precisely the images of the projective endpoints under the inverse Möbius transformation used below. Their nonvanishing is equivalent to the two endpoint coefficients being nonzero. $\square$

Lemma 3.2 (Even-form square-class injection). *Let $F \in \mathbb{Z}[X, Y]$ be homogeneous of even degree six, and fix $w \in \mathbb{Z} \setminus \{0\}$. Suppose*

$$F(a_i, b_i) = t_i w^2 \quad (i = 1, 2),$$

where t_1, t_2 are signed squarefree integers and $b_1 b_2 \neq 0$. If $a_1/b_1 = a_2/b_2$, then $t_1 = t_2$. The same conclusion holds after a fixed invertible fractional-linear transformation of the ratios.

Proof. Equality of the ratios gives $(a_2, b_2) = \lambda(a_1, b_1)$ for some $\lambda \in \mathbb{Q}^\times$. Hence

$$\frac{t_2}{t_1} = \frac{F(a_2, b_2)}{F(a_1, b_1)} = \lambda^6 = (\lambda^3)^2.$$

Two signed squarefree integers representing the same element of $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ are equal. An invertible fractional-linear transformation preserves equality of ratios. $\square$

4. A LOCALIZED LOG-FREE SQUAREFREE-VALUE PROPOSITION

We isolate the exact consequence of Stewart–Top needed in the proof. The statement includes the bounded positive witnesses that are present in the proof of their theorem, although not displayed in the definition of their unbounded counting function.

Proposition 4.1. *Let $F \in \mathbb{Z}[X, Y]$ be a squarefree binary form of degree $r = 6$, with nonzero leading and trailing coefficients. Suppose the largest degree of an irreducible factor over $\mathbb{Q}$ is $m \leq 5$. Fix integers A, B, M , with $M \geq 1$, and let $w \geq 1$ be maximal such that*

$$w^2 \mid F(a, b) \quad \text{whenever} \quad a \equiv A \pmod{M}, \quad b \equiv B \pmod{M}.$$

Then there are constants $c_0 > 0$ and $H_0 \geq 1$, depending only on the fixed data, such that every real $H \geq H_0$ admits at least $c_0 H^2$ distinct signed squarefree integers t for which

$$F(a, b) = t w^2$$

has a solution with $1 \leq a, b \leq H$, $a \equiv A \pmod{M}$, and $b \equiv B \pmod{M}$.

Proof. Apply [7, Theorem 1] with $k = 2$. Its hypothesis on the factor degrees is $m \leq 2k + 1 = 5$. The bounded-witness conclusion follows from the proof: the set constructed in [7, pp. 951–952] lies in the positive box $1 \leq a, b \leq H$ and in the fixed residue class. Equation (10) on p. 953 gives $\gg H^2$ eligible pairs. Among the finitely many possible values of $\gcd(a, b) \mid w$, retain the most frequent one and divide it out, as in (11). Equations (12)–(15), together with the uniform Thue-equation estimate of [6, Corollary 1], then bound the number of primitive representations of any one normalized value by a constant depending only on the fixed form and residue data. The paragraph following (15) therefore produces $\gg H^2$ distinct values of $F(a, b)$, not merely pairs. Since w is fixed and every retained value has the form $t w^2$, these yield the same number of distinct signed squarefree integers t .

The cited proof is first written for sufficiently large integral box parameters. For real H , use $\lfloor H \rfloor \geq H/2$ after enlarging H_0 , and shrink the constant by a factor of four. $\square$

Remark 4.2. The proof of Proposition 4.1 uses the distinct-value set obtained after the representation-multiplicity estimate, rather than identifying it with the earlier pair set. Also, a preliminary $\mathrm{SL}_2(\mathbb{Z})$ change of variables in the general proof is unnecessary once the two endpoint coefficients are nonzero. This is why Lemma 3.1 is imposed before the final congruence class is fixed.

5. THE CLASS-GROUP SPECIALIZATION INPUT

We use the following specialized form of the Kulkarni–Levin construction. It is a direct combination of their local-approximation lemma, thin-set height bound, specialization theorem, and the proof of their imaginary quadratic case [5, Lemmas 3.1–3.2, Theorems 1.3 and 4.1].

Fix a sufficiently large positive integer N_0 , coprime to the product M_0 of the bad primes of C_0 , such that

$$g(u) < 0 \quad \text{if} \quad \left| u + \frac{N_0}{M_0} \right| < 1. \quad (8)$$

This is possible because g is monic of odd degree. Put $\phi(P) = u(P) + N_0/M_0$. The specialization theorem supplies a fixed thin set $\Omega \subset \mathbb{Q}$ outside which the relevant torsion classes specialize to ideal classes.

Choose N_1 as in Lemma 3.1 and define

$$\psi(\alpha) = \frac{1 - N_1\alpha}{1 + N_1\alpha}, \quad \tau(q) = \psi^{-1}(q) = \frac{1 - q}{N_1(1 + q)}. \quad (9)$$

The proof of [5, Lemma 3.1] then gives fixed integers A, B, M —one may take $A = B = 1$ —such that positive integers in this single residue class satisfy all prescribed local conditions. The composition order is

$$q = (\psi \circ \phi)(P) = a/b, \quad \phi(P) = \tau(a/b), \quad u(P) = \tau(a/b) - N_0/M_0. \quad (10)$$

After clearing fixed denominators and removing fixed square factors, the branch equation can be written

$$g\left(\tau(X/Y) - \frac{N_0}{M_0}\right) = F(X, Y)R(X, Y)^2, \quad (11)$$

where $F \in \mathbb{Z}[X, Y]$ is a squarefree binary sextic and $R \in \mathbb{Q}(X, Y)$. The projective change of coordinate preserves separability and factor degrees, so F has factor-degree pattern 1, 1, 4. By Lemma 3.1, its leading and trailing coefficients are nonzero.

For the final residue class define w to be the maximal positive integer whose square divides every value in that class, exactly as in [7, p. 948]. All of $N_0, M_0, N_1, A, B, M, F, w, \Omega$ are now fixed; none depends on the later height parameter.

6. PROOF OF THE MAIN THEOREM

Proof of Theorem 1.1. Apply Proposition 4.1 to the binary sextic F from (11). There are $\gg H^2$ distinct signed squarefree integers t , each with a chosen witness (a_t, b_t) in the same positive box and residue class, such that

$$F(a_t, b_t) = tw^2. \quad (12)$$

The constants are uniform for every sufficiently large H .

For each t , set $q_t = \tau(a_t/b_t)$. A fixed fractional-linear map distorts multiplicative height by only a fixed constant, hence

$$H(q_t) \ll H. \quad (13)$$

By Lemma 3.2, the map $t \mapsto q_t$ is injective. The thin-set bound [5, Lemma 3.2] therefore shows that only $O(H)$ of the chosen parameters belong to Ω . Removing them leaves

$$\gg H^2 - O(H) \gg H^2 \quad (14)$$

distinct squarefree values.

For a surviving value, the fixed real neighborhood (8) and (11) imply $t < 0$, since the omitted factors and w^2 are squares. The field computation in the proof of [5, Theorem 1.2, equations (3.1)–(3.2)] gives

$$\mathbb{Q}(P_t) = \mathbb{Q}(\sqrt{F(a_t, b_t)}) = \mathbb{Q}(\sqrt{t}). \quad (15)$$

The local choices and the specialization theorem, together with Proposition 2.1, yield

$$\text{rk}_5 \text{Cl}(\mathbb{Q}(\sqrt{t})) \geq \text{rk}_5 \text{Jac}(C_0)(\mathbb{Q})_{\text{tors}} \geq 2. \quad (16)$$

Distinct signed squarefree t 's give distinct quadratic fields. In particular, the two conjugate points over one rational parameter are not double-counted: the argument counts squarefree values t , not geometric points.

Since F is fixed and has degree six, there is a constant $C_F > 0$ such that

$$|F(a, b)| \leq C_F H^6 \quad (1 \leq a, b \leq H).$$

The fundamental discriminant of $\mathbb{Q}(\sqrt{t})$ has absolute value at most $4|t|$. From (12) we obtain a fixed $C_\Delta > 0$ with

$$|\text{Disc}(\mathbb{Q}(\sqrt{t}))| \leq C_\Delta H^6. \quad (17)$$

Given sufficiently large X , take

$$H = \left\lfloor (X/C_\Delta)^{1/6} \right\rfloor.$$

Then $H \gg X^{1/6}$, all the surviving fields satisfy the discriminant cutoff X , and their number is $\gg H^2 \gg X^{1/3}$. This proves the stated estimate. $\square$

7. QUANTIFIERS, DEPENDENCIES, AND SCOPE

The order of choices in the proof is

$$\begin{aligned} C_0 &\implies (S, M_0, N_0, \phi, \Omega) \implies N_1 \\ &\implies (A, B, M, F, w, c_0, H_0, C_\Delta) \\ &\implies \forall H \geq H_0 \implies \forall X \geq X_0. \end{aligned} \tag{18}$$

Thus the residue class, the class-wise fixed square divisor, the thin set, all exceptional sets, and every implied constant are selected before H and X . This prevents a variable local condition or a height-dependent square factor from being hidden in the Vinogradov symbol.

Three source-level bookkeeping points are worth recording.

- (i) If a binary form is precomposed with a matrix L , the residue class preserving the original values is transported by L^{-1} . In the present argument the optional change is not used, because Lemma 3.1 makes both endpoint coefficients nonzero.
- (ii) The pair set in the proof of [7, Theorem 1] is not itself the distinct-value set. The passage to distinct values occurs only after the representation-multiplicity estimates (11)–(15).
- (iii) Two conjugate points can generate the same quadratic field. Counting distinct signed squarefree values t removes this multiplicity.

These observations do not change the exponent, but they are needed to keep the objects being counted aligned throughout the proof.

8. RELATED WORK AND NOVELTY BOUNDARY

The result closest to Theorem 1.1 is the unconditional estimate (2). Byeon’s $X^{1/4}$ theorem is log-free but has a smaller exponent [3]. The general Kulkarni–Levin corollary for $m = 5$ has exponent $1/5$ and a $(\log X)^{-2}$ factor [5, Corollary 1.4]. The conditional result of Chattopadhyay–Saikia has exponent $1/4 - \varepsilon$ for $p = 5$ [4].

The scoped search supporting this comparison included the exact notations $N_{5,2}^-(X)$ and $N^-(5^2; X)$, equivalent finite-abelian-group formulations, citation descendants of the principal sources, and current records in OpenAlex, Semantic Scholar, Crossref, and arXiv through 14 August 2026. It did not exhaust subscription databases, nonpublic manuscripts, unindexed theses, non-English literature, or work affected by indexing lag. Accordingly, the appropriate scholarly formulation is “to the best of our knowledge within the stated search scope,” rather than an absolute priority claim.

9. VERIFICATION AND REPRODUCIBILITY

The proof was assembled from frozen versions of the four principal sources [1, 5, 7, 6] and subjected to independent adversarial review. Exact rational arithmetic checks the model identity in Proposition 2.1, the forward curve

identity, both coordinate compositions, and the rejection of the incorrect inverse coefficient 1024, which would return $y/8$ instead of y .

In the accompanying project, the principal replay commands are

```
python3 mathematics/worker/no-log-blit-c0-rank-replay.py
python3 mathematics/worker/verify-route-audit.py
lean -q -t 0 mathematics/formal/NoLogSourceBridge.lean
lean -q -t 0 mathematics/formal/NoLogInference.lean
```

The two Lean files certify only the final conditional numerical assembly and elementary algebraic components. They do not formalize the Stewart–Top, Kulkarni–Levin, or Bartz–Levin–Thamminana inputs, nor the full unconditional analytic bridge. The complete theorem is therefore not claimed here as an end-to-end machine-verified result. The manuscript source and checks are bound to project commit `c9ca34210ee58d01c1f6fc5cd97f969534dbaa70` before the manuscript commit itself.

AI ASSISTANCE DISCLOSURE

DeepScientist, an AI-assisted research system, was used for structured literature discovery, extraction of proof obligations, symbolic-algebra replay, adversarial consistency checking, and preparation of the initial LaTeX draft. DeepScientist is not an author. Hao Zhang is the sole author, retains responsibility for every mathematical statement and citation, and should complete the conventional final human source and proof review required by the intended journal before submission.

ACKNOWLEDGEMENTS

The author thanks the authors of the cited works for making the constructions and source materials on which this argument builds available.

APPENDIX A. EXACT IDENTIFICATION OF THE TWO GENUS-TWO MODELS

For the sample $(t, u, z) = (2/3, -1/3, 25)$ in [1, Section 2], exact substitution gives

$$\begin{aligned} a &= \frac{2048}{243}, & a_0 &= -\frac{10240}{531441}, & a_2 &= \frac{20480}{59049}, \\ a_4 &= \frac{182272}{177147}, & a_6 &= \frac{31129600}{531441}. \end{aligned}$$

Let

$$P(x) = a(a_6x^6 + a_4x^4 + a_2x^2 + a_0).$$

Writing $D = 5x - 1$ and $N = 2(1 - 2x)$, the forward map (5) is certified by the coefficientwise identity

$$1024^2 D^6 f_0(N/D) = 59049^2 P(x). \quad (19)$$

Moreover,

$$5x_0 + 4 = \frac{6}{D}, \quad \frac{x_0 + 2}{5x_0 + 4} = x,$$

and the y -coordinate composition factor is

$$\frac{8192}{2187(5x_0 + 4)^3} \frac{59049}{1024D^3} = 1.$$

The old numerator 1024 in the first factor would give $1/8$, so the coefficient 8192 in (6) is forced. These are rational function identities on the dense affine opens where the denominators are nonzero; the induced maps extend to the smooth projective curves.

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